

Eco-Evolutionary Rescue and Basin Structure in a Roy-Style Predator–Prey Model

Mathbio research notes

May 29, 2026

Abstract

We study whether spatial structure can alter indirect evolutionary rescue in a predator–prey model with evolving prey defense. A fixed-defense spatial control did not produce robust predator persistence beyond the well-mixed reference. Introducing prey defense evolution in the well-mixed model increased the predator mortality stress tolerated by the system, consistent with indirect evolutionary rescue through a decline in defense frequency. In the spatial eco-evolutionary model, however, persistence was not organized by a single monotone stress threshold. Multi-horizon, continuation, and basin analyses showed path-dependent dynamics and bistability: at the same mortality stress, predator-persistent and predator-extinct outcomes were both reachable from different initial conditions. A focused basin-boundary scan showed that initial defense frequency and predator abundance influence basin entry. Subsequent ODE-PDE comparison showed that this basin dependence is primarily reaction-dominated: representative PDE solutions remained nearly homogeneous and matched ODE trajectories from the same initial means. These results suggest that spatially extended eco-evolutionary rescue should be analyzed in terms of reachable outcome basins rather than scalar persistence thresholds, while spatial-pattern-mediated rescue is not currently supported in this tested parameterization.

1 Introduction

Evolutionary rescue occurs when adaptive trait or genotype-frequency change allows a population to persist after environmental stress [Gomulkiewicz and Holt, 1995, Bell and Gonzalez, 2009, Bell, 2017]. In the present predator–prey setting, the relevant trait is prey defense. Higher defense reduces predation pressure, but it also carries ecological costs through lower prey growth and lower predator conversion opportunity. Indirect evolutionary rescue of the predator can occur when increased predator mortality lowers predator pressure, selection favors reduced prey defense, and the resulting lower-defense prey state improves predator recovery [Yamamichi and Miner, 2015].

This mechanism cannot be tested in a fixed-defense model. A fixed-defense spatial predator–prey model can ask whether spatial patterning alone creates predator persistence beyond a well-mixed reference, but it cannot test evolutionary rescue because the prey defense state does not change. The eco-evolutionary model therefore introduces a defense frequency q , with $q = 0$ representing fully undefended prey and $q = 1$ representing fully defended prey.

Comparing well-mixed and spatial models is useful because diffusion can alter local predator–prey overlap, transient recovery paths, and the basins of attraction of persistent or extinct states. Predator–prey systems are also a central setting in which rapid evolution and ecological dynamics can interact on similar timescales [Yoshida et al., 2003, Kasada et al., 2014]. A scalar mortality threshold m_c is a natural first diagnostic: below the threshold the predator persists, while above it the predator fails to persist. This diagnostic is appropriate when the stress response is monotone

and when persistence is controlled by one attracting outcome. It can fail in bistable systems, where persistent and extinct states are both reachable at the same stress.

This report distinguishes spatially extended dynamics from genuinely spatial-pattern-mediated dynamics. A PDE can preserve basin-dependent outcomes without those basins being generated by persistent spatial patterning. The main result is therefore two-part: the well-mixed eco-evolutionary model supports indirect evolutionary rescue, and the spatially extended model is not summarized by a single persistence threshold, but the current mechanism diagnosis is reaction-dominated homogeneous multistability rather than spatial-pattern-mediated rescue.

1.1 Relation to previous eco-evolutionary rescue models

This report follows the conceptual structure of indirect evolutionary rescue models in which ecological persistence depends on evolutionary change in another component of the community. Here the evolutionary coordinate is prey defense frequency. Predator mortality stress changes predator pressure; predator pressure changes selection on defense; and the resulting defense state feeds back on prey growth, prey palatability, and predator conversion.

The defense trade-off is intentionally minimal: defended prey have lower growth and lower vulnerability or conversion value than undefended prey. Such defense-growth and defense-palatability trade-offs are enough to create a feedback in which a stressed predator may recover indirectly when prey defense declines [Yamamichi and Miner, 2015]. The form of these trade-offs can qualitatively alter eco-evolutionary predator-prey dynamics [Kasada et al., 2014].

The present model is closest in spirit to indirect evolutionary rescue models in which predator mortality stress changes selection on prey defense. In the well-mixed case, the mechanism matches the qualitative logic proposed by Yamamichi and Miner [2015]: predator decline reduces predation pressure, lower predation pressure favors less-defended prey, and the resulting change in prey state improves predator persistence. The spatial extension studied here asks whether this mechanism remains describable by a mortality threshold once diffusion and spatial initial-condition dependence are introduced.

The contribution of the present report is mathematical and numerical rather than a final biological generalization. It shows that adding spatial dynamics changes the rescue question from a scalar threshold comparison to a basin-structured dynamical problem in the tested parameterization.

2 Model formulation

The eco-evolutionary variables are total prey density n , predator density w , prey defense frequency q , free space z , and predator mortality stress s . Free space is

$$z = \kappa^{-1} - n - w.$$

The defense trade-off functions are

$$r(q) = r_u(1 - q) + r_v q,$$

$$a(q) = a_u(1 - q) + a_v q,$$

$$b(q) = b_u(1 - q) + b_v q.$$

The well-mixed eco-evolutionary ODE is

$$\frac{dn}{dt} = n[r(q)z - \xi - a(q)w], \quad (1)$$

$$\frac{dw}{dt} = w[b(q)nz - (m + s) - \mu w], \quad (2)$$

$$\frac{dq}{dt} = \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \quad (3)$$

The spatial extension uses the same local reaction terms and adds diffusion:

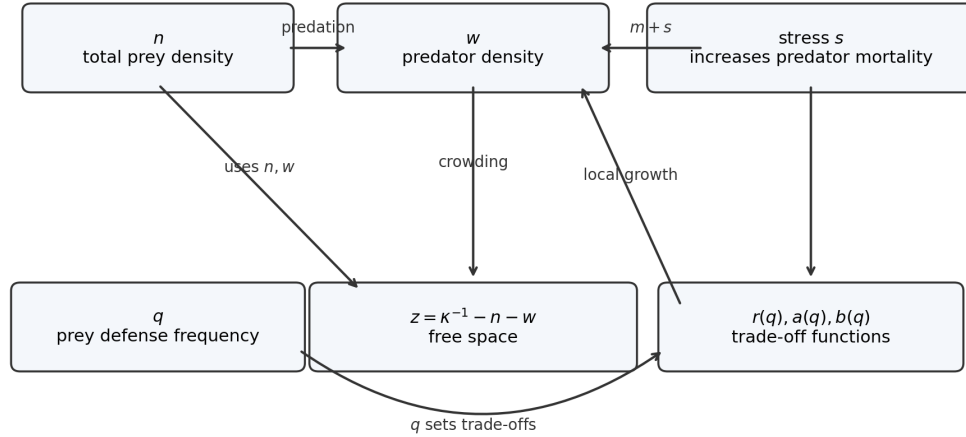
$$\partial_t n = D_n \nabla^2 n + n[r(q)z - \xi - a(q)w], \quad (4)$$

$$\partial_t w = D_w \nabla^2 w + w[b(q)nz - (m + s) - \mu w], \quad (5)$$

$$\partial_t q = D_q \nabla^2 q + \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \quad (6)$$

The spatial simulations use zero-flux Neumann boundary conditions.

Roy eco-evolutionary rescue model variables



ODE uses spatial means; PDE adds diffusion of n , w , and q with zero-flux boundaries.

Figure 1: Eco-evolutionary model variables and trade-off links. Arrows indicate that defense frequency q sets the growth, attack, and conversion functions $r(q)$, $a(q)$, and $b(q)$; prey and predator densities determine free space $z = \kappa^{-1} - n - w$; and mortality stress s increases predator mortality.

3 Numerical questions and diagnostic criteria

The first numerical question was whether prey defense evolution increases the predator mortality stress tolerated by the well-mixed system. This was measured by comparing the no-evolution and evolution thresholds

$$m_c^{ODE, \text{no evo}}, \quad m_c^{ODE, \text{evo}},$$

Table 1: Parameter values used in the eco-evolutionary tests. The conversion values $b_u = 0.08$ and $b_v = 0.02$ were used to obtain an interior defense-frequency baseline.

Symbol	Value	Description
κ	0.15	free-space scale
ξ	0.55	prey loss term
r_u	1	undefended prey growth
r_v	0.65	defended prey growth
a_u	1	undefended predation coefficient
a_v	0.35	defended predation coefficient
b_u	0.08	undefended conversion coefficient
b_v	0.02	defended conversion coefficient
m	0.1	baseline predator mortality
μ	0.2	predator density dependence
ν	0.05	evolutionary rate
D_n	0.01	prey diffusion
D_w	0.01	predator diffusion
D_q	0.005	defense-frequency diffusion

and their difference

$$\Delta_{evo}^{ODE} = m_c^{ODE, \text{evo}} - m_c^{ODE, \text{no evo}}.$$

A positive value indicates that prey evolution expands the predator persistence range in the well-mixed model.

The second numerical question was whether adding spatial diffusion changes the same rescue effect. A scalar PDE threshold was initially tested because it gives a direct comparison with the ODE threshold when persistence changes monotonically with stress. The later spatial diagnostics were needed because the apparent PDE boundary was sensitive to the tail window, integration horizon, and tail-slope rule. Multi-horizon analysis showed long transients near the apparent boundary, continuation experiments showed direction-dependent outcomes, and basin scans showed that different initial conditions can reach different outcomes at the same stress.

The outcome classes used in the spatial interpretation are scientific rather than procedural. A persistent outcome has a predator density tail that remains positive and steady enough to support persistence. An extinct outcome has predator density near zero. A transient outcome has not settled clearly into either class at the tested horizon. A bistable stress is one where both persistent and extinct outcomes are reachable from different initial states or continuation paths.

3.1 Numerical protocol

Well-mixed ODE thresholds were estimated by stress continuation and bisection, with predator persistence assessed from tail behavior. PDE scalar thresholds were used only as preliminary screening quantities because a scalar threshold is meaningful only if stress response is effectively monotone.

Spatial simulations used zero-flux boundaries and diffusion in n , w , and q . Initial PDE persistence calls were based on final-time and tail-window behavior, but sensitivity to tail window, integration horizon, and tail-slope rule motivated multi-horizon diagnostics. Continuation tests then used final states from neighboring stress values as initial states to test path dependence. Basin scans first varied broad initial-condition families and later varied q_0 and w_0/w_{baseline} directly.

PDE outcomes were interpreted as persistent, extinct, transient, or bistable using tail behavior, steady-state residual checks, and whether multiple outcomes were reachable at the same stress.

4 Fixed-defense spatial model as a control

Before introducing prey evolution, the fixed-defense spatial model was used as a control for spatial patterning alone. The question was whether spatial heterogeneity could create predator persistence beyond the well-mixed reference when prey defense was held fixed.

The fixed-defense spatial model did not produce robust ODE-extinct/PDE-persistent predator rescue. The strongest weak candidate occurred at $D_w/D_u = 150$, but further grid and mechanism checks classified it as a transient or numerical candidate rather than a robust spatial rescue mechanism. This control establishes that fixed spatial patterning alone was insufficient for predator rescue in the tested model.

5 Indirect evolutionary rescue in the well-mixed model

The well-mixed eco-evolutionary model gave the clean positive result. The no-evolution predator mortality stress threshold was

$$m_c^{ODE, \text{no evo}} = 0.069448242,$$

whereas the evolving-defense threshold was

$$m_c^{ODE, \text{evo}} = 0.16486816.$$

Thus

$$\Delta_{evo}^{ODE} = 0.095419922.$$

In the rescue window, defense frequency declined from approximately 0.6726 to approximately 0.3336.

Mechanistically, increased predator mortality reduces predator pressure. Lower predator pressure makes defense less favorable because the growth cost of defense is no longer offset by strong predation avoidance. As q declines, prey growth and predator conversion opportunity increase, allowing predator persistence over a larger stress interval. This is the cleanest evidence for indirect evolutionary rescue in the model sequence and follows the same qualitative feedback identified by [Yamamichi and Miner \[2015\]](#).

Table 2: Threshold quantities from the ODE rescue analysis and preliminary PDE screening. The PDE entries are early screening quantities; later diagnostics show that they should not be treated as final scalar thresholds for the spatial system.

Quantity	Value	Interpretation
$m_c^{ODE, \text{no evo}}$	0.0694482422	ODE reference
$m_c^{ODE, \text{evo}}$	0.164868164	ODE reference
Δ_{evo}^{ODE}	0.0954199219	ODE evolutionary rescue effect
representative $m_c^{PDE, \text{no evo}}$	0.06921875	historical screening quantity
representative $m_c^{PDE, \text{evo}}$	0.11765625	historical screening quantity

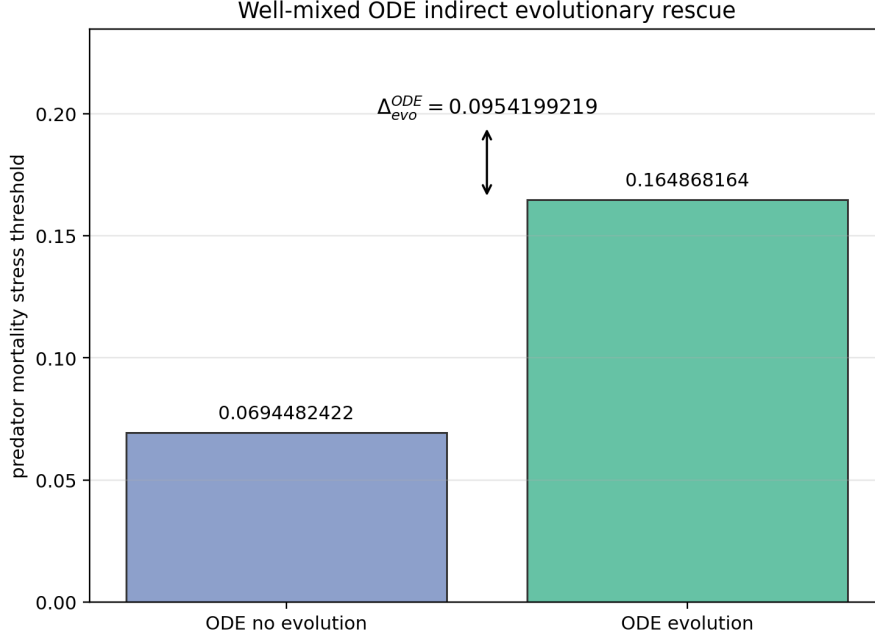


Figure 2: Well-mixed ODE threshold comparison. The no-evolution threshold is $m_c^{ODE, \text{no evo}} = 0.069448242$, the evolving-defense threshold is $m_c^{ODE, \text{evo}} = 0.16486816$, and the evolutionary rescue effect is $\Delta_{evo}^{ODE} = 0.095419922$.

6 Spatial eco-evolutionary model

The spatial eco-evolutionary model adds diffusion of n , w , and q to the same local eco-evolutionary reactions. The first scalar-threshold screen suggested that the PDE tolerated less mortality stress than the well-mixed evolving system. The representative screening values were $m_c^{PDE, \text{no evo}} = 0.06921875$ and $m_c^{PDE, \text{evo}} = 0.11765625$. Because $0.11765625 < 0.16486816$, this preliminary comparison suggested a lower PDE-evolution persistence boundary than the ODE-evolution threshold.

That comparison was useful as a warning signal, but it was not a stable final summary of the spatial system. The spatial model showed sensitivity to persistence criteria, long transients, and path-dependent outcomes. These features change the scientific question from “What is the PDE threshold?” to “Which persistent, extinct, or transient outcomes are reachable at a given stress?”

7 Evidence for path dependence and bistability

The spatial diagnostics proceeded through four conceptual phases. First, persistence classification was found to depend on the tail window and integration horizon. Second, longer multi-horizon analysis showed that several stresses near the apparent boundary remained transient rather than cleanly persistent or extinct. Third, continuation experiments showed direction-dependent outcomes: upward continuation could maintain a predator-persistent branch at stresses where downward continuation from high stress could remain extinct or transient. Finally, initial-condition basin scans confirmed that predator-persistent and predator-extinct outcomes can both be reached at the same stress.

Continuation data illustrate the path dependence. At stress 0.16486816, upward continuation produced a persistent steady outcome, whereas downward continuation produced an extinct steady

Diagnostics showing why a scalar PDE threshold is insufficient

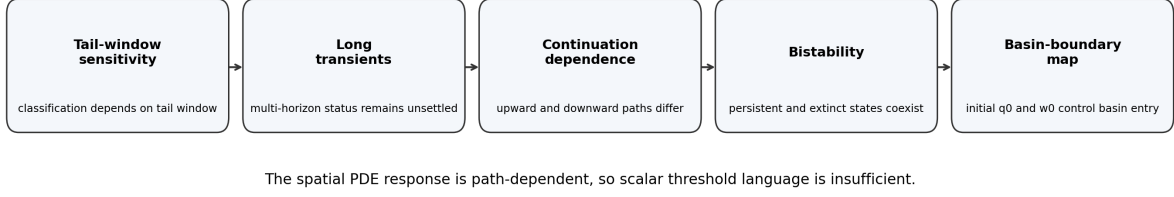


Figure 3: Diagnostic sequence showing why scalar PDE threshold language failed. Tail-window sensitivity, long transients, continuation dependence, and basin scans together showed that the spatial system is organized by path-dependent outcomes rather than one monotone persistence boundary.

outcome. At stress 0.175, the same contrast persisted. These paired outcomes show that the spatial model cannot be represented by one monotone persistence curve.

The broader basin-regime map identifies two types of stress regimes:

- mixed persistent/transient regimes at 0.141262205 and 0.15;
- bistable persistent/extinct regimes at 0.1584375, 0.16486816, and 0.175.

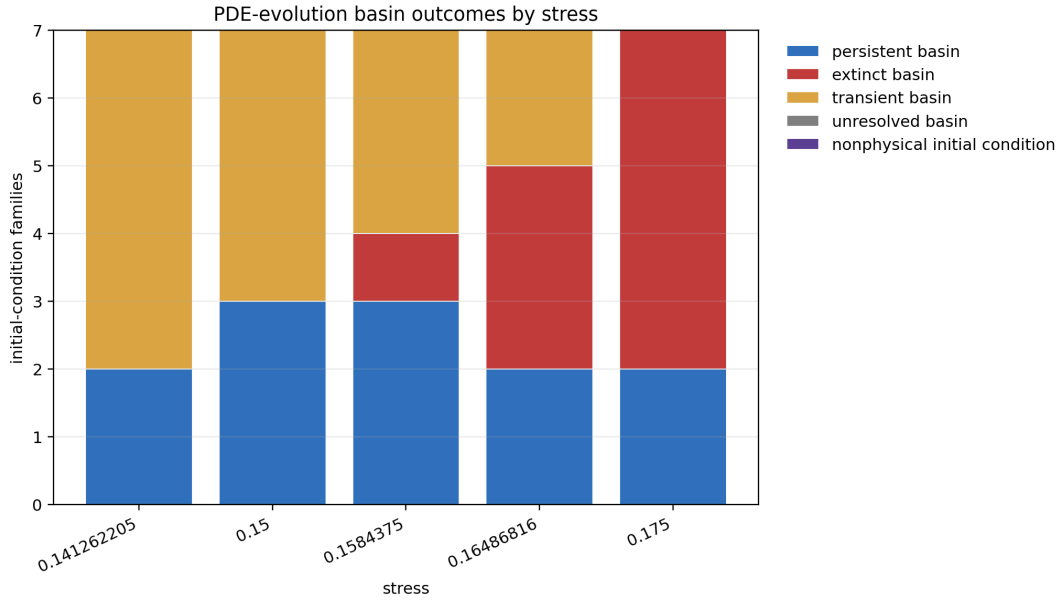


Figure 4: Basin outcomes across initial-condition families. Each stacked bar counts initial-condition families by basin label at a fixed stress; coexistence of persistent and extinct outcomes in the same bar indicates bistability.

Table 3: Stress-regime map from the bistability analysis.

Stress	Regime
0.141262205	persistent_transient_mixed
0.15	persistent_transient_mixed
0.1584375	bistable_persistent_extinct
0.16486816	bistable_persistent_extinct
0.175	bistable_persistent_extinct

8 Basin-boundary dependence on q_0 and w_0

The focused basin-boundary scan varied initial defense frequency q_0 and initial predator abundance scale w_0/w_{baseline} at two stresses inside the bistable interval. At stress 0.1584375, the q_0 - w_0 grid gave 20 persistent, 7 extinct, and 43 transient outcomes. At stress 0.16486816, it gave 14 persistent, 21 extinct, and 35 transient outcomes. No unresolved or nonphysical initial-condition outcomes were recorded in these two focused grids.

The scan shows that basin entry is two-dimensional. Low-to-intermediate q_0 values tend to favor predator-persistent outcomes, so the initial defense state is a major basin-entry coordinate. Predator initial abundance modulates the outcome within that q_0 range. Transient outcomes remain common, which means the separatrix is not yet fully resolved at the tested horizon.

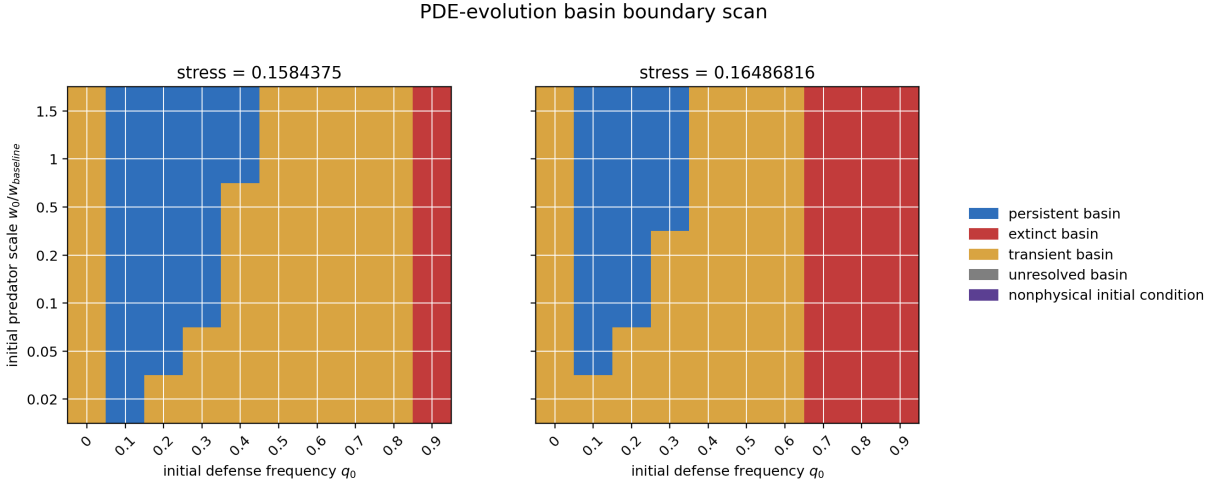


Figure 5: Basin-boundary scan over initial defense frequency q_0 and predator abundance scale w_0/w_{baseline} at stresses $s = 0.1584375$ and $s = 0.16486816$. Colors encode basin labels for persistent, extinct, transient, unresolved, or nonphysical outcomes.

Table 4: Basin-boundary counts from the q_0 - w_0 scan.

Stress	Persistent	Extinct	Transient	Unresolved	Nonphysical	Regime
0.1584375	20	7	43	0	0	bistable_persistent_extinct
0.16486816	14	21	35	0	0	bistable_persistent_extinct

Table 4 highlights the stress-dependent basin balance. At $s = 0.1584375$, persistent and extinct

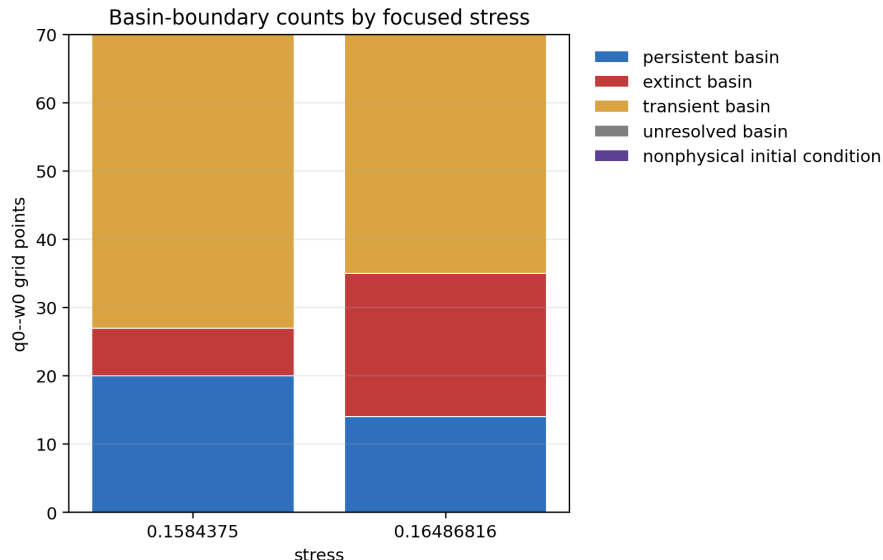


Figure 6: Outcome counts in the focused q_0 – w_0 basin-boundary scan. Transient outcomes remain common at both stresses, so the basin boundary is interpretable but the separatrix is not fully resolved at the tested horizon.

basins both occur, but transient outcomes dominate the grid. At $s = 0.16486816$, extinct outcomes become more common, yet persistent outcomes remain reachable. Therefore the basin boundary is interpretable but not sharply resolved at the current horizon.

9 Mechanism diagnosis: homogeneous reaction dynamics versus spatial patterning

The basin-boundary scan established that the spatially extended PDE preserves multiple reachable outcomes, but it did not by itself identify whether spatial patterning generates those outcomes. A subsequent matched ODE-PDE mechanism test compared representative PDE spatial means with ODE trajectories from the same initial means, quantified representative spatial variance, compared ODE and PDE basin labels on the same q_0 – w_0 grid, and ran targeted perturbation-sensitivity tests.

The mechanism diagnosis was **reaction_dominated_homogeneous_multistability**. Representative ODE-PDE classifications agreed in 3 of 3 cases, basin-grid labels agreed for 90% of 140 q_0 – w_0 points, all 14 disagreements involved transient labels, direct persistent/extinct disagreements were zero, final spatial CV values were very small, and perturbation tests produced no outcome changes. Thus the current evidence indicates that the observed PDE basin structure is mostly inherited from the homogeneous eco-evolutionary reaction dynamics rather than generated by persistent spatial patterning.

The follow-up ODE analysis therefore treats the homogeneous reaction system as the next object of study. The representative ODE trajectories track predator density, defense frequency, the selection gradient

$$(r_v - r_u)z - (a_v - a_u)w,$$

and the predator growth factor

$$b(q)nz - (m + s) - \mu w.$$

The ODE q_0 – w_0 basin map reproduces persistent, extinct, and transient regions on the same focused grid, and the numerical equilibrium scan gives parameter-specific evidence for multiple homogeneous attractors. These diagnostics support the conditional interpretation that basin-dependent rescue is controlled by homogeneous eco-evolutionary feedbacks under the tested trade-off values.

The equilibrium scan suggests that the persistent homogeneous branch compensates increased predator mortality by lowering the equilibrium defense frequency q^* , while n^* and w^* remain close to the same positive values over the tested stress range. This provides a reaction-level explanation for the rescue mechanism and for why the spatial PDE can inherit basin-dependent outcomes without persistent spatial patterning.

10 Discussion

The fixed-defense control shows that spatial patterning alone was insufficient for robust predator rescue in the tested Roy-style model. The well-mixed eco-evolutionary model then shows that evolutionary change in prey defense can rescue predators under mortality stress, consistent with indirect evolutionary rescue theory [Yamamichi and Miner, 2015]. The spatially extended eco-evolutionary model preserves basin-dependent outcomes, but the current mechanism audit indicates that those basins are primarily inherited from homogeneous reaction dynamics.

The PDE result is not a simple amplification or suppression of the well-mixed rescue threshold; it is a spatially extended expression of basin-dependent dynamics. This changes what “spatial effect on rescue” means. Rather than asking whether the PDE threshold is higher or lower than the ODE threshold, the relevant question is which outcomes are reachable at each stress, how the basins depend on initial ecological and evolutionary state, and whether those basins are generated by homogeneous reactions or by persistent spatial patterning.

The basin-boundary results indicate that initial defense frequency and predator abundance are important coordinates for basin entry. Low-to-intermediate defense frequencies more often reach predator persistence, while high-defense states and some low-predator states more often lead to extinction or unresolved transients. Because transient outcomes remain common, the current basin map should be read as a resolved first layer of basin structure, not as a fully refined separatrix.

10.1 Strength of current claims

Strongly supported in the tested parameterization:

- well-mixed prey defense evolution increases predator mortality tolerance;
- fixed-defense spatial patterning did not produce robust predator rescue;
- the spatially extended eco-evolutionary model preserves basin-dependent outcomes in the tested stress range;
- the current mechanism diagnosis is reaction-dominated homogeneous multistability rather than spatial-pattern-mediated rescue.

Moderately supported:

- initial defense frequency q_0 and predator abundance w_0 influence basin entry.

Unresolved:

- the exact separatrix in q_0 – w_0 space;

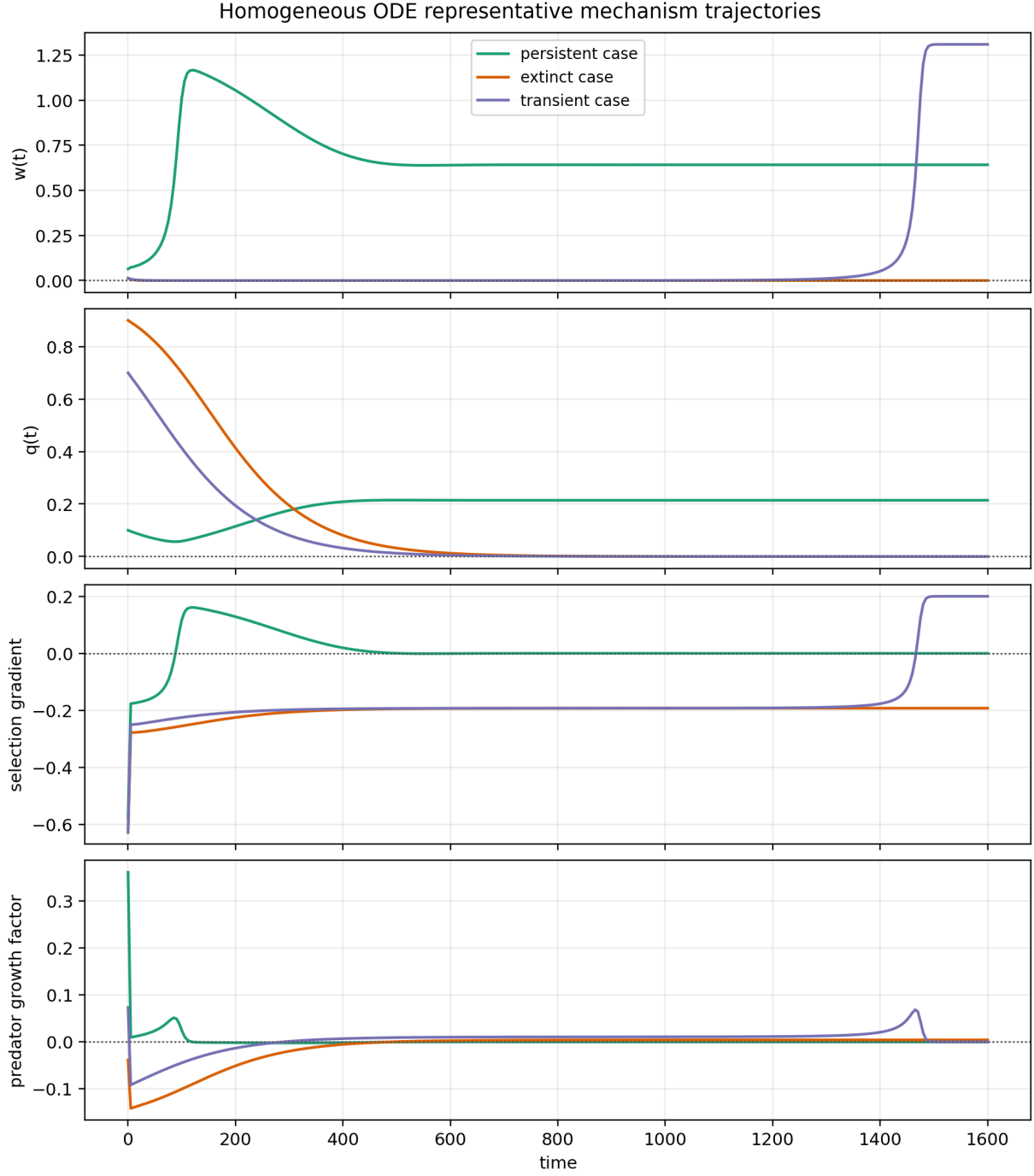


Figure 7: Representative homogeneous ODE trajectories for predator density w , defense frequency q , selection gradient, and predator growth factor. The cases use the same initial means as the representative PDE comparisons.

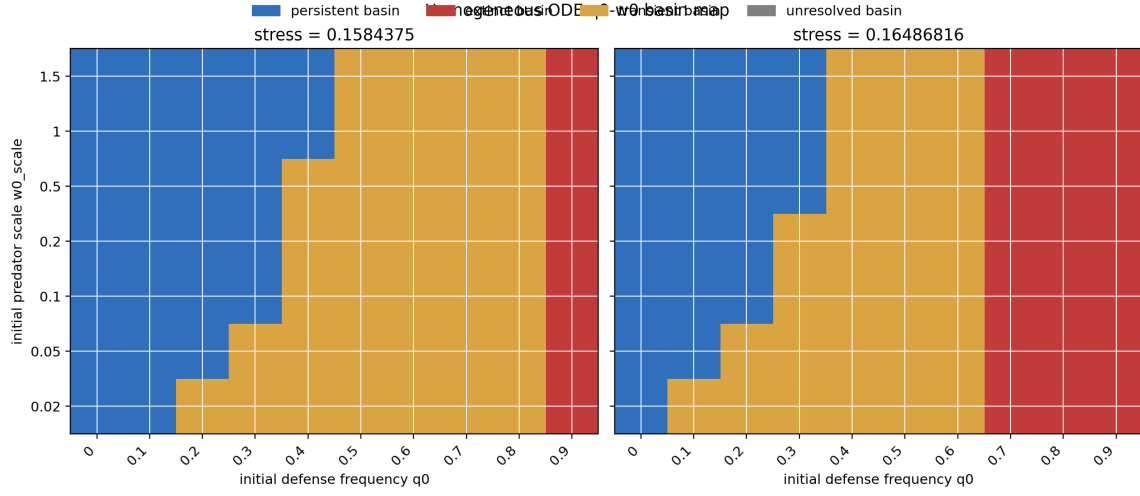


Figure 8: Homogeneous ODE basin map on the same q_0 - w_0 grid used in the ODE-PDE basin comparison.

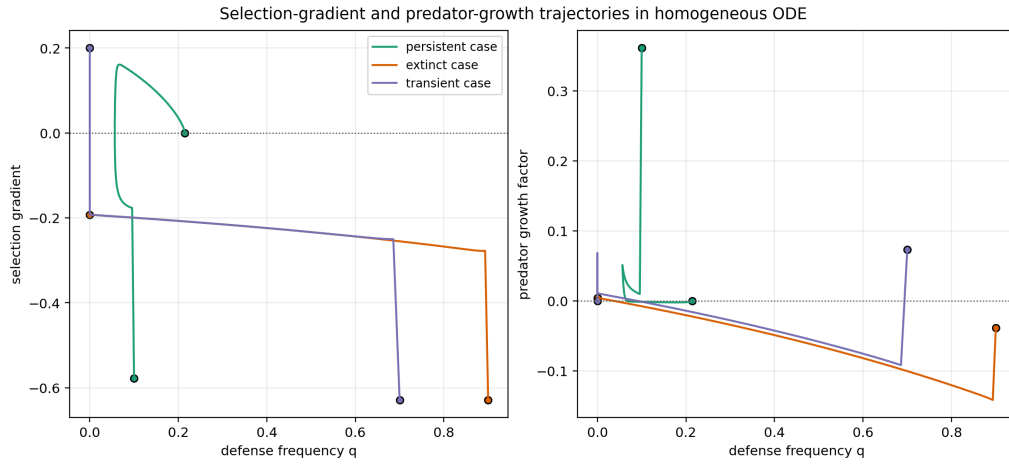


Figure 9: Representative ODE relationship between defense frequency q , the selection gradient on defense, and the predator growth factor.

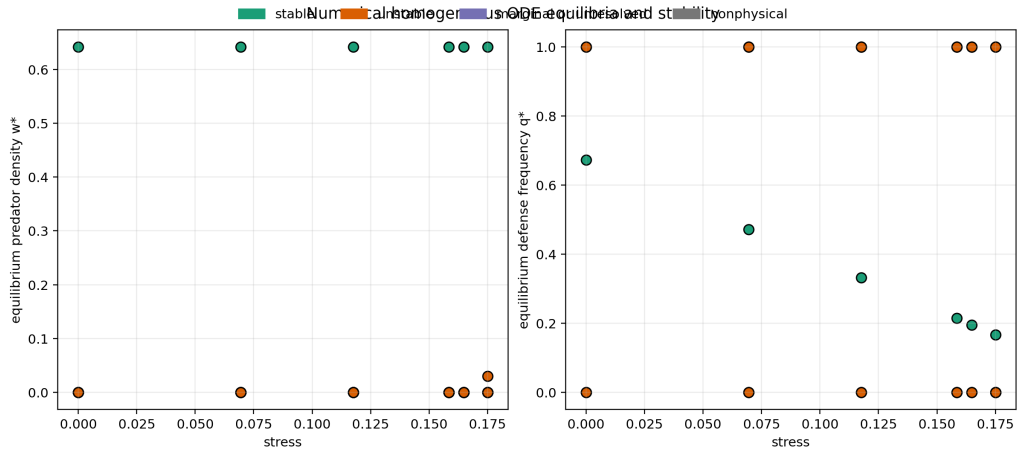


Figure 10: Numerical homogeneous equilibria across targeted stress values. Marker color indicates stability classification from finite-difference Jacobian eigenvalues.

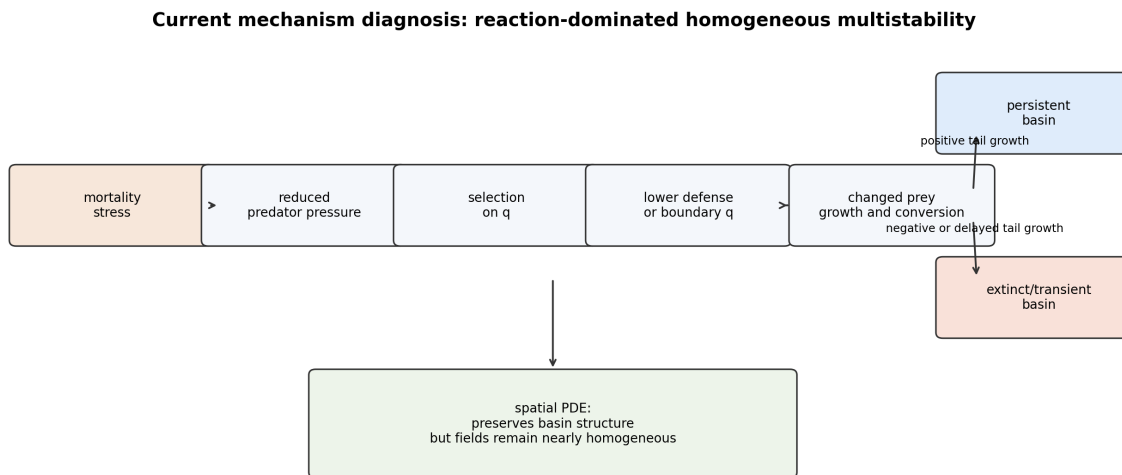


Figure 11: Updated mechanism diagram. Mortality stress alters predator pressure, which changes selection on prey defense and can move the homogeneous reaction system into predator-persistent, extinct, or transient basins. In the current tests, the spatial PDE preserves this basin structure while representative solutions remain nearly homogeneous.

- sensitivity to diffusion coefficients;
- generality across trade-off forms and parameterizations;
- analytical characterization of the homogeneous equilibrium and basin structure;
- biological calibration.

11 Limitations and future work

The current analysis has several limitations. It uses one trade-off parameterization, with $b_u = 0.08$ and $b_v = 0.02$. Transient outcomes remain common near the basin boundary. The q_0 – w_0 basin boundary has not yet been adaptively refined. Sensitivity to diffusion coefficients has not yet been tested after basin mapping. The model has also not been externally calibrated to a particular biological system.

The next numerical task is to test how the homogeneous basin structure depends on the defense trade-off parameters and to refine the analytical equilibrium and stability interpretation. Further PDE basin refinement may be useful later, but it should follow the current mechanism correction rather than replace it.

In the well-mixed system, prey defense evolution supports indirect evolutionary rescue. In the spatially extended system, the same eco-evolutionary reaction mechanism does not reduce to a single persistence threshold. Instead, the PDE preserves basin-dependent outcomes, with basin entry controlled in part by initial prey defense frequency and predator abundance. Persistent spatial patterning is not currently supported as the mechanism in this parameterization.

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